AUTONOMOUS VIDEO PIPELINE CONTRACT

End-to-End Video Production Architecture

A rigorous engineering specification for transforming raw content (Topics, Text, URLs) into broadcast-ready, sub-frame synchronized, 1080×1920 vertical motion graphics.

Engine: **Remotion 4 (Chromium Headless)**
AI Provider: **Codex CLI / Deep Researcher**
Audio: **Kokoro TTS + Whisper STT + Procedural SFX**
Output: **1080×1920 @ 30 FPS**

RULE 01
Audio-First Timing
Visuals are never constructed before narration audio exists. Scene bounds derive strictly from spoken words.

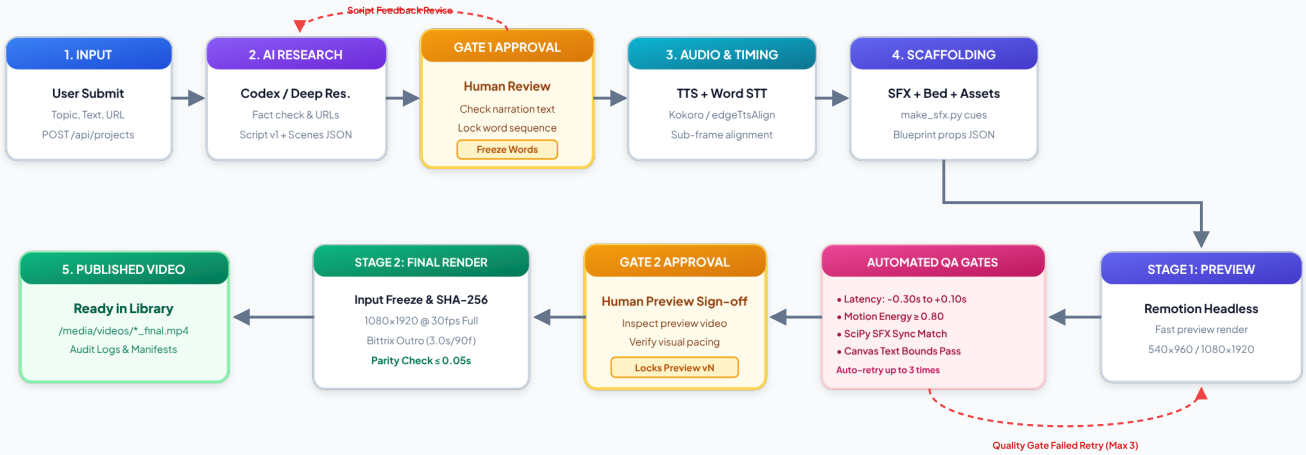
RULE 02
Zero Gaps / Overlaps
Scenes tile the entire timeline seamlessly. Total video duration matches audio length to within 1 frame.

RULE 03
Motion Energy ≥ 0.80
Every scene maintains continuous, purposeful camera zoom, orbiting light, and drift. No static frames permitted.

RULE 04
Two-Stage Render Gate
Preview render passes automated latency, motion, and sync checks before final render input freeze.

Complete End-to-End Pipeline Workflow Diagram

FIGURE 1.0: CANONICAL FLOW FROM USER SUBMISSION TO PUBLISHED VIDEO LIBRARY



RULE 27 MANDATE **Bittrix Technologies 3-Second Outro:** Every generated animation MUST conclude with an authentic branded outro scene rendering `remotion/public/bittrix-logo.mp4` (exactly 90 frames @ 30 FPS / 180 frames @ 60 FPS) with brand CTA narration.

Upstream Execution: Ingestion, Research & Audio Alignment

The Animation Bittrix pipeline operates through strict state machine transitions with explicit human-in-the-loop validation barriers. Below are the execution contracts for Phases 1 through 4.

PHASE 01

Content Submission & Deep AI Research

`POST /api/projects`

Accepts raw input (`topic` , `text` , or `url`). The backend invokes the Codex CLI / Deep Researcher skill to ground all assertions in primary, verified academic and institutional sources. Shallow summaries are rejected.

- **Script Contract:** Exactly 1 short, impactful sentence per scene. Scene 1 is always an attention hook.
- **Blueprint Scene JSON:** Produces narration, visual blueprint tags, camera cues, and color palette.
- **Source Verification:** Mandates open URLs, fact-checking notes, and token audit records.
- **Database State:** Project stage transitions to `script_review` with script version 1.

GATE 01

Human Script Approval Barrier (Narration Freeze)

`POST /api/projects/:id/scripts/approve`

A reviewer audits narration phrasing, pacing, and visual intentions in the Web Dashboard. No audio generation or render job can proceed without this explicit cryptographic and database sign-off.

- **Narration Immutability:** Once approved, the exact wording is frozen. Visual adjustments cannot alter spoken words.
- **Revision Loop:** If changes are requested (`/scripts/revise`), a new monotonic version is spawned.

PHASE 02

Speech Synthesis & Sub-Frame Alignment

`Kokoro TTS + Whisper STT`

Synthesizes high-fidelity speech for each scene narration sentence. Audio files are converted to broadcast MP3 via `ffmpeg` and measured with `ffprobe` to determine precise frame counts.

- **Dynamic Scene Timing:** Scene start = word_0 - (5 to 8 frames). Scene end = word_N + short hold.
- **Word-Level Captions:** Sub-frame word timestamps drive active kinetic highlighting with 0.00s offset.
- **Zero Gaps/Overlaps:** Audio files are concatenated into a master track. Scene durations tile the track perfectly.
- **No Time-Stretching:** If voice exceeds target limits, Kokoro natural speed is incremented; no artificial warps.

PHASE 03

Procedural SFX & Acoustic Bed Synthesis

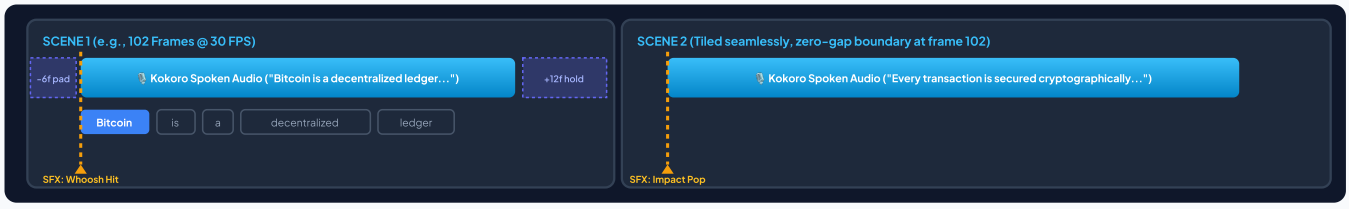
`scripts/make_sfx.py`

Every on-screen object entrance, kinetic typography hit, and camera movement is scored with a procedural, license-free sound effect aligned to the exact transient hit frame.

- **Sound Roles:** Synthesizes deterministic pop, click, whoosh, slide, riser, impact, sparkle, chime, and ding.
- **Transient Shift:** Compensates for sound rise time so the auditory transient peaks on the visual frame.
- **Continuous Bed:** Generates a procedural background bed with automatic speech ducking (-14dB).
- **Cue Manifest:** Writes `.cues.json` containing frame-by-frame sound triggers and volume envelopes.

Figure 2.0: Sub-Frame Audio & Video Timeline Synchronization

AUDIO-DRIVEN SCENE DURATION & SFX CUE TRANSIENT ALIGNMENT FORMULA



Downstream Execution: Motion, Quality Gates & Final Release

The rendering pipeline strictly bifurcates into Stage 1 (Fast Preview & Automated Verification) and Stage 2 (Locked Input Production Render).

PHASE 04

Visual Scaffolding & Continuous Motion Rules

`Remotion 4 + React Canvas`

Constructs React composition props adhering to vertical 9:16 layout standards. Embeds kinetic typography, vector schematics, 3D models, and dynamic particle systems.

- **Continuous Motion Rule:** Every scene must maintain continuous slow zoom, orbiting glow, and fluid drift. Motion Energy ≥ 0.80.
- **Zero Meta-Clutter:** Forbids debug text, terminal windows, code chrome, or placeholder counters.
- **Mobile Safe Zone:** Keeps text elements within strict vertical margins (top 15%, bottom 20% clear for social UI).
- **Mandatory Outro:** Renders official Bittrix outro (`bittrix-logo.mp4`) during final 90 frames (3.0s).

PHASE 05

Stage 1: Headless Preview Render & Quality Gates

`remotion-host.mjs + qualityGate.mjs`

Renders a versioned preview (`<topic>_preview_vN.mp4`) and executes an automated, 8-point pass/fail QA validation suite before the preview is eligible for human review.

- **Latency Validation:** Scene lag must remain between -0.30s and +0.10s. Caption lag between -0.10s and +0.20s.
- **Acoustic Sync Check:** `verify_sync.py` verifies SFX audio transients against visual hit frames using SciPy cross-correlation.
- **Canvas Overlap Gate:** Checks text bounding boxes to ensure no overlapping typography or clipped safe-area zones.
- **Retry Guardrail:** If any check fails, system auto-adjusts layout/timing and retries (maximum 3 attempts).

GATE 02

Human Preview Review & Approval Barrier

`POST /api/projects/:id/previews/approve`

The reviewer watches the completed preview video in the Frontend player. Approval must target the exact preview version ID and completed render job before final production is unlocked.

- **Version Pinning:** Prevents race conditions by pinning the exact preview artifact that was inspected.
- **Visual Feedback Loop:** If rejected, visual revisions spawn a new blueprint version without rewriting audio.

PHASE 06

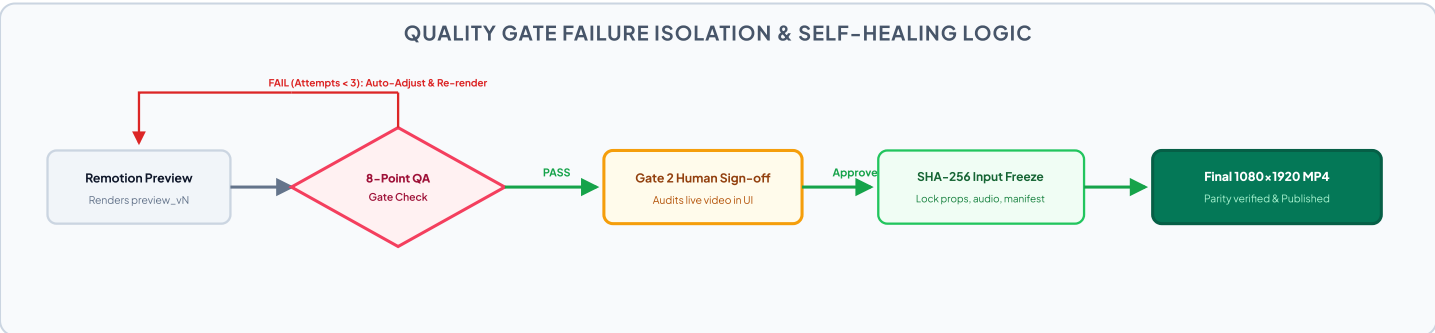
Stage 2: Input Freeze & Full 1080×1920 Production Render

`POST /api/projects/:id/final-render`

Freezes all inputs with SHA-256 checksums. Executes the full-resolution final render. Validates strict preview-to-final parity before delivering to the Video Library.

- **Input Freeze:** Locks audio clips, cue manifests, and JSON props. No assets are refetched or recomputed.
- **Parity Verification:** Confirms final render scene starts and caption starts match preview within 0.05 seconds.
- **Library Publishing:** Saves final deliverable to `/media/videos/:id_final.mp4` with full duration and metadata.
- **Audit Persistence:** Stores permanent run records in database tables `videos` and `agent_runs`.

Figure 3.0: Automated Two-Stage QA Decision Tree



Quality Gates, Metrics & Execution Reference

The Animation Bittrix pipeline relies on mathematical and empirical validation at every step. Below are the quantitative quality thresholds and API/CLI controls.

Automated Quality Gate Standards

METRIC / CRITERION	TOLERANCE THRESHOLD	LEVEL
Motion Energy (Per Scene)	≥ 0.80 (Target ≥ 0.90)	Strict
Scene Audio Alignment	-0.30s to +0.10s lag	Strict
Word Caption Alignment	-0.10s to +0.20s lag	Strict
Cumulative Timing Drift	0.00s across entire timeline	Strict
Audio/Video Duration Match	≤ 1 frame difference	Strict
SFX Acoustic Peak Sync	SciPy cross-correlation peak ≤ 1f	Verified
Canvas Text Safe Area	Top 15%, Bottom 20% clear	Verified
Bittrix Outro Video	Exactly 90f (3.0s) closing	Mandatory

System Architecture & Components

LAYER	TECHNOLOGY / MODULE	PRIMARY FUNCTION
Frontend	React 18 + Vite + Tailwind	Review UI, Gate controls, WebSockets
Backend	Node.js Express (ESM)	State machine, REST API, orchestration
Database	PostgreSQL 15+	Audit trails, script versions, jobs
AI Provider	Codex CLI / Deep Researcher	Scholarly research, scene blueprints
TTS / STT	Kokoro TTS + Whisper STT	Voiceover synthesis, word timestamps
SFX Engine	scripts/make_sfx.py (NumPy)	Procedural sound transient generator
Renderer	Remotion 4 (Headless Chromium)	React composition to 1080×1920 MP4
Sync QA	scripts/verify_sync.py (SciPy)	Audio/visual cross-correlation check

REST API & CLI Execution Interface

METHOD & PATH	PAYLOAD / ARGUMENTS	STATE EFFECT & BEHAVIORAL RESULT
`POST /api/projects`	`{ input_content, input_type, title? }`	Initializes project, triggers Codex research, returns with Script v1 stored.
`POST /api/projects/:id/scripts/approve`	`{ script_id, version_number, approved_by }`	**GATE 1:** Enforces script freeze. Transitions stage to `script_approved`.
`POST /api/projects/:id/scripts/revise`	`{ feedback, change_narration? }`	Creates next monotonic script version. Preserves narration unless explicitly requested.
`POST /api/projects/:id/previews/render`	`{ script_version, visual_feedback? }`	Synthesizes audio, builds SFX, renders preview MP4, and executes QA validation suite.
`POST /api/projects/:id/previews/approve`	`{ render_job_id, preview_version, approved_by }`	**GATE 2:** Locks exact preview version. Transitions stage to `preview_approved`.
`POST /api/projects/:id/final-render`	`None (Uses locked preview inputs)`	Freezes SHA-256 inputs, renders final 1080×1920 MP4, publishes to video library.
`./scripts/pipeline-cli.sh`	`"<Topic>" [DurationSec] [FPS]`	Autonomous standalone CLI pipeline runner: research, audio, props, render, and QA in one command.

Auditing & Reproducibility: Every video generation creates immutable audit records in `backend/media/render-data/` including the exact JSON props, SFX cue manifests (`.cues.json`), asset manifests, and token usage logs. Any deliverable can be deterministically re-rendered frame-by-frame.